

Why Go to the s -Domain?

A Worked Comparison on a Two-Mass, Two-Spring System

The Laplace transform is usually introduced as a table of rules. What it's *for* is easy to lose: turning a differential equation into an algebra problem. For a single spring-mass-damper, that trade barely shows – solving the ODE directly and solving it via Laplace take about the same amount of work. This handout uses a slightly bigger system – two masses coupled by two springs, no damping – where the trade becomes obvious.

We solve the *same* system, for the *same* input, twice: once entirely in the time domain, once via the s -domain round trip. Both routes must agree – and they do, exactly. What's worth watching is *how much work each route costs*, and how that cost changes as the system gets more complicated.

The System

Two masses $m_1 = m_2 = 1$, two springs $k_1 = k_2 = 1$, no friction anywhere. An input force $u(t)$ drives m_1 through spring k_1 ; m_1 is coupled to m_2 through spring k_2 . The output is x_2 , the position of the second mass. The equations of motion are:

$$\ddot{x}_1 + 2x_1 = u + x_2 \qquad \ddot{x}_2 + x_2 = x_1$$

We'll drive the system with a unit step, $u(t) = 1$ for $t \geq 0$, starting from rest: $x_1(0) = \dot{x}_1(0) = x_2(0) = \dot{x}_2(0) = 0$.

Route 1: Solve Directly in the Time Domain

Step 1 – Eliminate x_1

We want a single equation in x_2 alone. From the second equation, $x_1 = \ddot{x}_2 + x_2$. Differentiating *twice* to get $\ddot{x}_1$:

$$\ddot{x}_1 = x_2^{(4)} + \ddot{x}_2$$

Substitute both into the first equation:

$$(x_2^{(4)} + \ddot{x}_2) + 2(\ddot{x}_2 + x_2) = u + x_2 \implies x_2^{(4)} + 3\ddot{x}_2 + x_2 = u$$

For our unit step ($u = 1$):

$$x_2^{(4)} + 3\ddot{x}_2 + x_2 = 1$$

This already cost us two extra differentiations just to get a solvable single-variable equation – work that scales up fast with every extra mass added to the chain.

Step 2 – Homogeneous solution

Characteristic equation: $r^4 + 3r^2 + 1 = 0$. Substitute $y = r^2$:

$$y^2 + 3y + 1 = 0 \implies y = \frac{-3 \pm \sqrt{5}}{2}$$

Both roots are negative, so r^2 is negative both times: the four roots are purely imaginary, $r = \pm i\omega_1, \pm i\omega_2$, with

$$\omega_1 = \sqrt{\frac{3-\sqrt{5}}{2}} \approx 0.618 \quad \omega_2 = \sqrt{\frac{3+\sqrt{5}}{2}} \approx 1.618$$

(No damping anywhere in the system, so no decay – purely imaginary poles are exactly what we should expect.) The homogeneous solution is

$$x_{2,h}(t) = A \cos \omega_1 t + B \sin \omega_1 t + C \cos \omega_2 t + D \sin \omega_2 t$$

Step 3 – Particular solution

Constant forcing $\Rightarrow$ try a constant $x_{2,p} = K$: all derivatives vanish, leaving $K = 1$. General solution:

$$x_2(t) = 1 + A \cos \omega_1 t + B \sin \omega_1 t + C \cos \omega_2 t + D \sin \omega_2 t$$

Step 4 – The hidden cost: four initial conditions on x_2

The physics only handed us $x_2(0) = 0$ and $\dot{x}_2(0) = 0$. A 4th-order equation needs *four*. The other two are not given – we have to go back to the *original coupled equations* to derive them:

$$\ddot{x}_2 = x_1 - x_2 \implies \ddot{x}_2(0) = x_1(0) - x_2(0) = 0$$

$$\ddot{\dot{x}}_2 = \dot{x}_1 - \dot{x}_2 \implies \ddot{\dot{x}}_2(0) = \dot{x}_1(0) - \dot{x}_2(0) = 0$$

This step has no counterpart in the s -domain route – there, “zero initial conditions” is a one-line assumption and nothing more is asked of it. Here, every extra order of the ODE means hunting down one more derivative’s worth of initial condition from the original system.

Step 5 – Solve for A, B, C, D

Differentiate the general solution three times and evaluate all four at $t = 0$:

$$x_2(0) = 1 + A + C = 0$$

$$\dot{x}_2(0) = B\omega_1 + D\omega_2 = 0$$

$$\ddot{x}_2(0) = -A\omega_1^2 - C\omega_2^2 = 0$$

$$\ddot{\dot{x}}_2(0) = -B\omega_1^3 - D\omega_2^3 = 0$$

The second and fourth equations involve only B, D : solving them together forces $B\omega_1(\omega_1^2 - \omega_2^2) = 0$, and since $\omega_1 \neq \omega_2$,

$$B = 0, \quad D = 0$$

The first and third equations involve only A, C :

$$A + C = -1, \quad A\omega_1^2 + C\omega_2^2 = 0$$

Solving this 2×2 system:

$$A = \frac{-5 - 3\sqrt{5}}{10} \approx -1.1708 \quad C = \frac{3\sqrt{5} - 5}{10} \approx 0.1708$$

Step 6 – Final answer

$$x_2(t) = 1 - 1.1708 \cos(0.618 t) + 0.1708 \cos(1.618 t)$$

Six steps, two of them (eliminating x_1 , then hunting down hidden initial conditions) pure book-keeping that exists only because we're working in the time domain.

Route 2: The s -Domain Round Trip

Step 1 – Transform both equations

Zero initial conditions turn every derivative into a power of s , with *no extra work*: $\mathcal{L}\{\ddot{x}\} = s^2 X(s)$, full stop – no matter how high the order.

$$(s^2 + 2) X_1(s) = U(s) + X_2(s) \qquad (s^2 + 1) X_2(s) = X_1(s)$$

Step 2 – Eliminate X_1 – pure algebra

Solve the first equation for X_1 and substitute into the second:

$$\begin{aligned} X_1(s) &= \frac{U(s) + X_2(s)}{s^2 + 2} \quad \implies \quad (s^2 + 1)(s^2 + 2) X_2(s) = U(s) + X_2(s) \\ X_2(s) \left[(s^2 + 1)(s^2 + 2) - 1 \right] &= U(s) \quad \implies \quad X_2(s) [s^4 + 3s^2 + 1] = U(s) \\ X_2(s) &= \frac{U(s)}{s^4 + 3s^2 + 1} \end{aligned}$$

Same denominator as Route 1's characteristic polynomial – reached with two lines of algebra, no differentiation, no elimination-by-substitution gymnastics.

Step 3 – Plug in the input

Unit step: $U(s) = 1/s$.

$$X_2(s) = \frac{1}{s(s^4 + 3s^2 + 1)} = \frac{1}{s(s^2 + \omega_1^2)(s^2 + \omega_2^2)}$$

Step 4 – Partial fractions

$$X_2(s) = \frac{1}{s} + \frac{B_1 s + C_1}{s^2 + \omega_1^2} + \frac{B_2 s + C_2}{s^2 + \omega_2^2}$$

Clearing denominators and matching powers of s (a purely mechanical, table-driven procedure) gives $C_1 = C_2 = 0$ automatically, and

$$B_1 = \frac{-5 - 3\sqrt{5}}{10} \approx -1.1708 \qquad B_2 = \frac{3\sqrt{5} - 5}{10} \approx 0.1708$$

– the *exact same numbers* as A and C from Route 1, arrived at by matching polynomial coefficients instead of solving a linear system built from four derivatives.

Step 5 – Inverse transform, term by term

Using only $\mathcal{L}^{-1}\{1/s\} = 1$ and $\mathcal{L}^{-1}\{s/(s^2 + \omega^2)\} = \cos \omega t$ from the table:

$$x_2(t) = 1 + B_1 \cos \omega_1 t + B_2 \cos \omega_2 t$$

Step 6 – Final answer

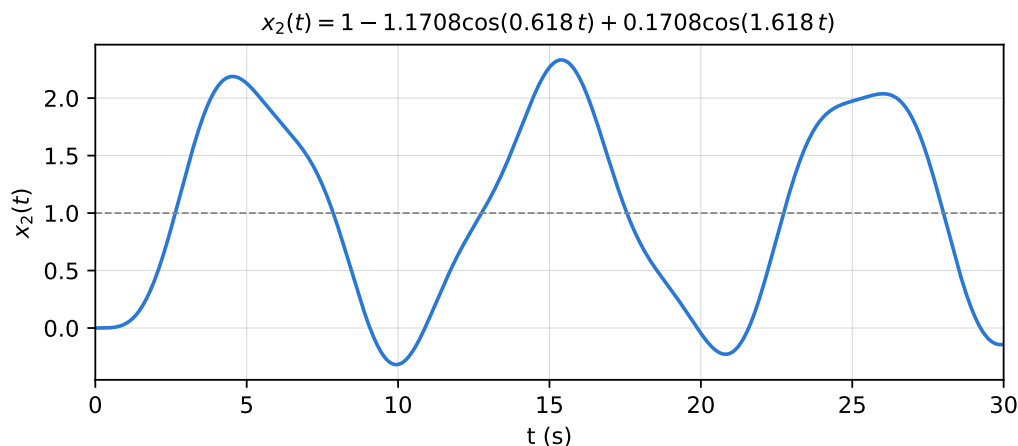
$$x_2(t) = 1 - 1.1708 \cos(0.618 t) + 0.1708 \cos(1.618 t)$$

Identical to Route 1 – as it must be, since both routes solve the same equations. (For what it's worth: $0.618\dots$ and $1.618\dots$ are $1/\varphi$ and φ , the golden ratio – a fact of this particular characteristic polynomial, $s^4 + 3s^2 + 1$, not a general feature of coupled spring–mass systems.)

The Comparison

	Time domain	<i>s</i> -domain
Turning the ODEs into something solvable	Eliminate x_1 by substitution, requiring two extra differentiations	One line of algebra per equation
Handling initial conditions	Must <i>derive</i> $\ddot{x}_2(0)$, $\ddot{x}_1(0)$ from the original coupled system – one extra derivation per extra order	Built into the transform; zero I.C.'s cost nothing, nonzero I.C.'s cost one extra term per derivative
Finding the unknown constants	Solve a 4×4 linear system built from evaluating derivatives at $t = 0$	Match polynomial coefficients (partial fractions) – no derivatives involved
Scaling to a 3-mass system, or adding damping	Every step above gets harder: more differentiations, more hidden I.C.'s to hunt down, a bigger linear system	Same three steps (transform $\rightarrow$ algebra $\rightarrow$ partial fractions) – just bigger polynomials

The point isn't that the *s*-domain is a trick. It's that differentiation becomes multiplication by s , and initial conditions get absorbed into the transform itself instead of requiring a fresh derivation every time the model gets one order bigger. For a single spring-mass-damper that trade is barely worth noticing. For a coupled, higher-order system, it's the difference between a page of algebra and a page of calculus.



Both routes agree exactly: an undamped system oscillates forever around its steady-state value of 1, beating between the two natural frequencies $\omega_1 \approx 0.618$ and $\omega_2 \approx 1.618$ rad/s.